

Assignment 02 – Solutions

General Physics II (PHY-108) | Summer 2026 | Faculty: RUH

QUESTION 01

(A) (i) For an empty (air-filled) parallel-plate capacitor:

$$C = \frac{\epsilon_0 A}{d}, \quad \epsilon_0 = 8.85 \times 10^{-12} \text{ F/m}$$

(ii) It stores **electric potential energy** in the electric field between the plates, $U = \frac{1}{2}CV^2 = \frac{q^2}{2C}$.

(iii) If V is too high, the field $E = V/d$ exceeds the dielectric strength of the medium. The air (or dielectric) between the plates gets ionized and becomes conducting $\Rightarrow$ **dielectric breakdown**: a spark jumps across, the capacitor discharges suddenly and may be permanently damaged.

(iv) Since $C = \kappa\epsilon_0 A/d$:

1. Increase the plate area A .
2. Decrease the plate separation d .
3. Insert a dielectric of $\kappa > 1$ between the plates.

(B) C_1 and C_2 are in **parallel** (both between A and B); C_4 and C_5 are in **parallel** (both between C and D). These two groups are in **series** with C_3 , and the whole chain is in **parallel** with C_6 (which is directly across A – D).

$$C_{12} = C_1 + C_2 = 12.0 + 5.30 = 17.3 \text{ } \mu\text{F}$$

$$C_{45} = C_4 + C_5 = 17.0 + 9.30 = 26.3 \text{ } \mu\text{F}$$

$$\frac{1}{C_s} = \frac{1}{17.3} + \frac{1}{4.50} + \frac{1}{26.3} = 0.05780 + 0.22222 + 0.03802 = 0.31805$$

$$C_s = 3.144 \text{ } \mu\text{F}$$

$$C_{eq} = C_s + C_6 = 3.144 + 10.0 = \underline{\underline{13.1 \text{ } \mu\text{F}}}$$

(C) Given: $C_1 = C_6 = 3.00 \text{ } \mu\text{F}$, $C_3 = C_5 = 4.00 \text{ } \mu\text{F}$, $C_2 = C_4 = 2.00 \text{ } \mu\text{F}$, $V = 20.0 \text{ V}$.

C_3 and C_5 are in series; that pair is in parallel with C_2 and C_4 . This whole group is in series with the parallel pair $C_1 \parallel C_6$.

(i)

$$C_{35} = \frac{C_3 C_5}{C_3 + C_5} = \frac{4 \times 4}{8} = 2.00 \text{ } \mu\text{F}$$

$$C_{2345} = C_2 + C_4 + C_{35} = 2.00 + 2.00 + 2.00 = 6.00 \text{ } \mu\text{F}$$

$$C_{16} = C_1 + C_6 = 3.00 + 3.00 = 6.00 \text{ } \mu\text{F}$$

$$C_{eq} = \frac{6.00 \times 6.00}{6.00 + 6.00} = \underline{\underline{3.00 \text{ } \mu\text{F}}}$$

$$q = C_{eq} V = 3.00 \times 20.0 = \underline{\underline{60.0 \text{ } \mu\text{C}}}$$

(ii) C_{16} and C_{2345} are in series $\Rightarrow$ each carries $q = 60.0 \text{ } \mu\text{C}$.

$$V_{16} = \frac{60.0}{6.00} = 10.0 \text{ V} \Rightarrow \underline{\underline{V_1 = 10.0 \text{ V}}}, \quad q_1 = C_1 V_1 = 3.00 \times 10.0 = \underline{\underline{30.0 \text{ } \mu\text{C}}}$$

(iii) $V_{2345} = \frac{60.0}{6.00} = 10.0 \text{ V}$, and C_2 is directly across this group:

$$\underline{\underline{V_2 = 10.0 \text{ V}}}, \quad q_2 = C_2 V_2 = 2.00 \times 10.0 = \underline{\underline{20.0 \text{ } \mu\text{C}}}$$

(iv) The series pair C_{35} is also across 10.0 V:

$$q_3 = q_{35} = C_{35}(10.0) = 2.00 \times 10.0 = \underline{20.0 \mu C}, \quad V_3 = \frac{q_3}{C_3} = \frac{20.0}{4.00} = \underline{5.00 \text{ V}}$$

QUESTION 02

(A) (i) Conventional current is taken in the direction of flow of **positive** charge, i.e. from the + terminal to the – terminal through the external circuit. Electrons actually drift in the opposite direction.

(ii) An **ideal** emf device has no internal resistance, so its terminal potential difference always equals its emf ($V = \varepsilon$). A **real** emf device has internal resistance r , so $V = \varepsilon - ir$, which falls as the current increases.

(iii) **Resistivity does not change.** ρ is a property of the *material* (and temperature) only – it does not depend on geometry. Only the *resistance* changes:

$$R = \frac{\rho L}{A} = \frac{\rho L}{\pi r^2} \longrightarrow R' = \frac{\rho L}{\pi (r/2)^2} = 4R$$

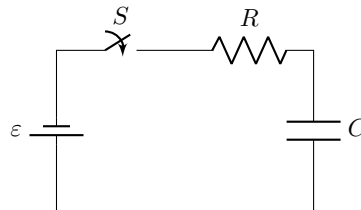
So R becomes 4 times larger, while ρ stays the same.

(B) (i) Current–voltage relation: $\boxed{V = iR}$ (Ohm's law), i.e. $i = V/R$.

Plot (1) is the **resistor**: the graph is a *straight line through the origin*, so $R = V/i$ is constant and the same for both polarities. The resistor is an **ohmic** device – the number of charge carriers is fixed, so $i \propto V$.

Plot (2) is the **diode**: the graph is *non-linear* and not symmetric. It is a **non-ohmic** device. A diode is a p – n junction: in forward bias the depletion region shrinks and current rises steeply after a threshold voltage; in reverse bias the depletion region widens and almost no current flows. Hence R is not constant – it depends on both the magnitude and the polarity of V .

(ii) Simple RC circuit:



The behaviour is fixed by the **capacitive time constant** $\tau = RC$:

$$\text{charging: } q = C\varepsilon \left(1 - e^{-t/RC}\right), \quad i = \frac{\varepsilon}{R} e^{-t/RC} \quad \text{discharging: } q = q_0 e^{-t/RC}$$

τ is the time for the charge to reach 63% of its final value (or fall to 37%).

(C) (i) *True value* (no voltmeter): ε , r , R_1 , R_2 in series,

$$i = \frac{\varepsilon}{r + R_1 + R_2} = \frac{3.0}{100 + 250 + 300} = \frac{3.0}{650} = 4.615 \text{ mA}$$

$$V_1 = iR_1 = (4.615 \times 10^{-3})(250) = 1.1538 \text{ V}$$

Measured value: the voltmeter R_V is in parallel with R_1 ,

$$R'_1 = \frac{R_1 R_V}{R_1 + R_V} = \frac{250 \times 5000}{5250} = 238.10 \Omega$$

$$i' = \frac{3.0}{100 + 238.10 + 300} = \frac{3.0}{638.10} = 4.702 \text{ mA}, \quad V'_1 = i' R'_1 = (4.702 \times 10^{-3})(238.10) = 1.1194 \text{ V}$$

$$\% \text{ error} = \frac{V_1' - V_1}{V_1} \times 100 = \frac{1.1194 - 1.1538}{1.1538} \times 100 = \underline{-2.99 \%}$$

i.e. the voltmeter reads about 3.0% **lower** than the true potential difference.

(ii) $R = 15.0 \text{ k}\Omega$, $\varepsilon = 12.0 \text{ V}$, $V_C = 5.00 \text{ V}$ at $t = 1.30 \mu\text{s}$.

$$V_C = \varepsilon \left(1 - e^{-t/\tau}\right) \Rightarrow e^{-t/\tau} = 1 - \frac{5.00}{12.0} = 0.5833$$

$$\frac{t}{\tau} = \ln\left(\frac{12.0}{7.00}\right) = 0.5390 \Rightarrow \tau = \frac{1.30 \times 10^{-6}}{0.5390} = \underline{2.41 \mu\text{s}}$$

$$C = \frac{\tau}{R} = \frac{2.41 \times 10^{-6}}{15.0 \times 10^3} = 1.61 \times 10^{-10} \text{ F} = \underline{161 \text{ pF}}$$

QUESTION 03

(A) (i) *Outside* the magnet the field lines run from the **north pole to the south pole**; *inside* the magnet they run from south to north, so every line is a closed loop.

Properties of magnetic field lines:

- They form continuous **closed loops** (no beginning or end – no magnetic monopole).
- They **never intersect** each other.
- The tangent at any point gives the direction of $\vec{B}$ at that point.
- Where the lines are **denser**, the field is **stronger** (strongest near the poles).

(ii) **Magnetic force** is the force experienced by a moving charge, or by a current-carrying conductor, placed in a magnetic field:

$$\vec{F}_B = q\vec{v} \times \vec{B} \quad \text{and} \quad \vec{F}_B = i\vec{L} \times \vec{B}, \quad F = iLB \sin \phi$$

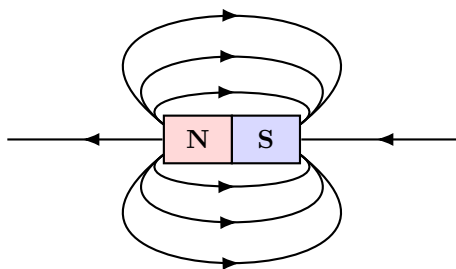
Direction – right-hand rule: point the fingers of the right hand along the current $\vec{L}$ (or velocity $\vec{v}$), curl them towards $\vec{B}$; the extended **thumb** gives the direction of $\vec{F}_B$. For a negative charge (electron) the force is opposite to this.

(iii) **Hall effect:** when a current-carrying conductor (strip) is placed in a magnetic field perpendicular to the current, the moving charge carriers are pushed sideways by the magnetic force and pile up on one edge. This sets up a transverse potential difference V_H (Hall voltage) across the strip, which grows until the electric force balances the magnetic force:

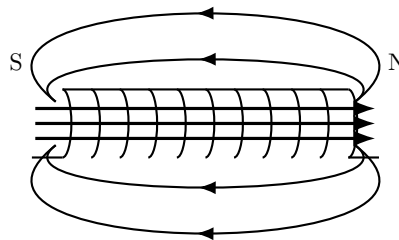
$$eE = ev_d B \Rightarrow n = \frac{Bi}{V_H l e}$$

It tells us the **sign** and the **number density** n of the charge carriers.

(B)



(i) Field of a bar magnet



(ii) Field of a solenoid ($B = \mu_0 ni$ inside)

Inside the solenoid the field is uniform and parallel to the axis; outside it is very weak. A solenoid behaves exactly like a bar magnet.

(C) (i) Take $\hat{i}$ along the electron's motion ($+x$), $\hat{j}$ upward ($+y$), $\hat{k}$ out of the page ($+z$).

Speed after acceleration through V_1 :

$$eV_1 = \frac{1}{2}mv^2 \Rightarrow v = \sqrt{\frac{2eV_1}{m}} = \sqrt{\frac{2(1.60 \times 10^{-19})(1.00 \times 10^3)}{9.11 \times 10^{-31}}} = 1.88 \times 10^7 \text{ m/s}$$

Electric field between the plates:

$$E = \frac{V_2}{d} = \frac{100}{20.0 \times 10^{-3}} = 5.00 \times 10^3 \text{ V/m}$$

The lower plate is at the lower potential, so $\vec{E}$ points **downward** ($-\hat{j}$). The electron has negative charge, so the electric force on it is **upward**, $\vec{F}_E = eE\hat{j}$.

For a straight-line path, the magnetic force must be equal and opposite (downward):

$$evB = eE \Rightarrow B = \frac{E}{v} = \frac{5.00 \times 10^3}{1.88 \times 10^7} = 2.67 \times 10^{-4} \text{ T}$$

With $\vec{F}_B = -e(\vec{v} \times \vec{B})$ and $\vec{v} = v\hat{i}$, a downward force requires $\vec{B}$ along $-\hat{k}$ (into the page):

$$\boxed{\vec{B} = -(2.67 \times 10^{-4} \text{ T}) \hat{k}} \quad (\text{into the page})$$

(ii) Period of the circular motion:

$$T = \frac{2\pi r}{v} = \frac{2\pi(5.29 \times 10^{-11})}{2.19 \times 10^6} = 1.52 \times 10^{-16} \text{ s}$$

Equivalent current and area:

$$i = \frac{e}{T} = \frac{1.60 \times 10^{-19}}{1.52 \times 10^{-16}} = 1.05 \times 10^{-3} \text{ A}, \quad A = \pi r^2 = \pi(5.29 \times 10^{-11})^2 = 8.79 \times 10^{-21} \text{ m}^2$$

Magnetic dipole moment:

$$\mu = iA = (1.05 \times 10^{-3})(8.79 \times 10^{-21}) = 9.28 \times 10^{-24} \text{ A} \cdot \text{m}^2$$

Torque is maximum when $\vec{\mu} \perp \vec{B}$ ($\sin \theta = 1$):

$$\tau_{max} = \mu B = (9.28 \times 10^{-24})(7.10 \times 10^{-3}) = \underline{\underline{6.59 \times 10^{-26} \text{ N} \cdot \text{m}}}$$

QUESTION 04

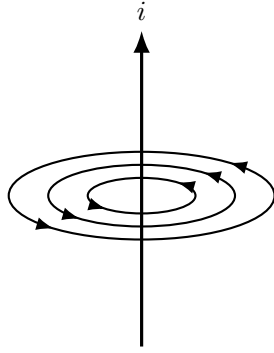
(A) (i) **Ampere's law:** the line integral of $\vec{B}$ around any closed loop equals μ_0 times the net current enclosed by that loop:

$$\oint \vec{B} \cdot d\vec{s} = \mu_0 i_{enc}$$

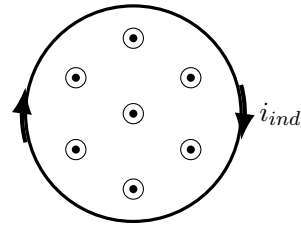
(ii) An **Amperian loop** is an imaginary closed curve drawn in the field region, used to apply Ampere's law. It is chosen to match the symmetry of the problem (a circle for a straight wire, a rectangle for a solenoid) so that B is either constant and parallel to $d\vec{s}$, or perpendicular to it – which makes the integral easy to evaluate.

(iii) i_{enc} is the **algebraic (net) sum** of the currents passing through the surface bounded by the loop. Curl the right hand along the chosen direction of integration; the thumb then points in the **positive** direction. Currents in that direction are taken $+$, currents opposite are taken $-$. Currents *outside* the loop are not included.

(B)



(i) $\vec{B}$ around a vertical wire
(counter-clockwise seen from above)



(ii) $\vec{B}$ out of page and increasing
 $\Rightarrow i_{ind}$ is **clockwise**

(i) By the right-hand rule (thumb up along i), the field lines are concentric circles around the wire, directed counter-clockwise when viewed from above.

(ii) By **Lenz's law**, the outward flux is increasing, so the induced current must oppose it by producing a field *into* the page inside the loop $\Rightarrow i_{ind}$ flows **clockwise**.

(C) (i) Wires: 1 at $(0, a)$, 2 at (a, a) , 3 at $(a, 0)$, 4 at $(0, 0)$, with $a = 8.50 \text{ cm} = 0.0850 \text{ m}$, $i = 15.0 \text{ A}$. All currents are parallel (out of page) $\Rightarrow$ all three wires **attract** wire 1.

$$\frac{F}{L} = \frac{\mu_0 i^2}{2\pi d}$$

For $d = a$ (wires 2 and 4):

$$\frac{F_{12}}{L} = \frac{(4\pi \times 10^{-7})(15.0)^2}{2\pi(0.0850)} = \frac{(2 \times 10^{-7})(225)}{0.0850} = 5.294 \times 10^{-4} \text{ N/m}$$

For $d = a\sqrt{2}$ (wire 3, along the diagonal at -45°):

$$\frac{F_{13}}{L} = \frac{5.294 \times 10^{-4}}{\sqrt{2}} = 3.743 \times 10^{-4} \text{ N/m}$$

Resolving (attraction $\Rightarrow$ each force points from wire 1 towards the other wire):

from 2 ($+x$): $(+5.294 \times 10^{-4}, 0)$

from 4 ($-y$): $(0, -5.294 \times 10^{-4})$

from 3 (-45°): $(3.743 \times 10^{-4} \cos 45^\circ, -3.743 \times 10^{-4} \sin 45^\circ) = (2.647 \times 10^{-4}, -2.647 \times 10^{-4})$

$$\frac{F_x}{L} = 7.94 \times 10^{-4} \text{ N/m}, \quad \frac{F_y}{L} = -7.94 \times 10^{-4} \text{ N/m}$$

$$\boxed{\frac{\vec{F}}{L} = (7.94 \times 10^{-4} \text{ N/m}) \hat{i} - (7.94 \times 10^{-4} \text{ N/m}) \hat{j}}$$

Magnitude $= 1.12 \times 10^{-3} \text{ N/m}$, directed at -45° (towards wire 3).

(ii) For a toroid, $B = \frac{\mu_0 N i}{2\pi r}$ with $N = 500$, $i = 0.800 \text{ A}$.

At the inner radius $r = 15.0 \text{ cm} = 0.150 \text{ m}$:

$$B = \frac{(4\pi \times 10^{-7})(500)(0.800)}{2\pi(0.150)} = \frac{(2 \times 10^{-7})(400)}{0.150} = \underline{\underline{5.33 \times 10^{-4} \text{ T}}}$$

At the outer radius $r = 15.0 + 5.00 = 20.0 \text{ cm} = 0.200 \text{ m}$:

$$B = \frac{(2 \times 10^{-7})(400)}{0.200} = \underline{\underline{4.00 \times 10^{-4} \text{ T}}}$$

(The field is not uniform – it falls off as $1/r$ across the cross section.)

QUESTION 05

(A) (i) An emf is induced whenever the **magnetic flux Φ_B through the loop changes** with time – by changing B , changing the area A , or changing the orientation θ of the loop.

Mutual inductance M between two coils depends on:

- the number of turns N_1, N_2 of the two coils,
- the size, shape and cross-sectional area of the coils,
- the distance between them and their relative orientation,
- the permeability μ of the core / medium between them.

(M does *not* depend on the current.)

(ii) **Faraday's law:** the induced emf equals the negative rate of change of magnetic flux,

$$\varepsilon = -N \frac{d\Phi_B}{dt}$$

Magnetic flux $\Phi_B = \int \vec{B} \cdot d\vec{A} = BA \cos \theta$ is a measure of the number of field lines passing through a surface. Unit: weber ($\text{Wb} = \text{T} \cdot \text{m}^2$).

(iii) **Lenz's law:** the induced current in a closed loop flows in such a direction that the magnetic field it produces **opposes the change in flux** that caused it. (This is the origin of the minus sign in Faraday's law; it is a statement of energy conservation.)

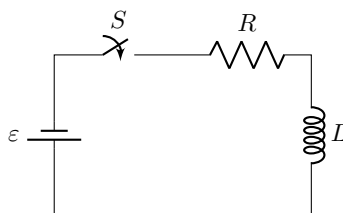
Induced current is the current driven around the closed loop by the induced emf, $i_{ind} = \varepsilon/R$.

(iv) **Eddy currents:** when a *solid/bulk* conductor is placed in a changing magnetic flux, the induced currents cannot be confined to a wire – they swirl in closed loops inside the body of the conductor. They dissipate energy as heat ($i^2 R$) and oppose the motion (magnetic braking / damping). In transformer cores they are reduced by using thin laminated sheets.

(v) **Self-induction:** when the current in a coil changes, its own magnetic flux changes, and this induces an emf in the *same* coil which opposes the change of current:

$$\varepsilon_L = -L \frac{di}{dt}, \quad L = \frac{N\Phi_B}{i}$$

(B) (i) Simple RL circuit:



(ii) The behaviour is fixed by the **inductive time constant $\tau_L = L/R$** :

$$\text{rise: } i = \frac{\varepsilon}{R} \left(1 - e^{-t/\tau_L}\right) \quad \text{decay: } i = i_0 e^{-t/\tau_L}$$

τ_L is the time for the current to reach 63% of its final value ε/R (or fall to 37%). The inductor opposes the *change* of current, so the current cannot jump instantly.

(C) (i) The coil radius (1.8 cm) is larger than the solenoid radius (1.6 cm), so the flux is limited by the **solenoid's** cross-section.

$$n = 220 \text{ turns/cm} = 2.20 \times 10^4 \text{ turns/m}, \quad R_s = \frac{3.2}{2} = 1.6 \text{ cm} = 0.016 \text{ m}$$

$$A = \pi R_s^2 = \pi(0.016)^2 = 8.04 \times 10^{-4} \text{ m}^2$$

$$B = \mu_0 n i = (4\pi \times 10^{-7})(2.20 \times 10^4)(1.5) = 4.15 \times 10^{-2} \text{ T}$$

$$\Phi_B = BA = (4.15 \times 10^{-2})(8.04 \times 10^{-4}) = 3.34 \times 10^{-5} \text{ Wb}$$

Since the current drops to zero, $\Delta\Phi_B = 3.34 \times 10^{-5} \text{ Wb}$ in $\Delta t = 25 \text{ ms}$:

$$|\varepsilon| = N \frac{\Delta\Phi_B}{\Delta t} = \frac{(120)(3.34 \times 10^{-5})}{25 \times 10^{-3}} = 0.160 \text{ V}$$

$$i_{ind} = \frac{\varepsilon}{R} = \frac{0.160}{5.3} = 0.0302 \text{ A} = \underline{\mathbf{30 \text{ mA}}}$$

(ii) From graph *b*: before and after the field is set up, $i = 1.5 \text{ mA}$; during $t = 10 \text{ s}$ to 20 s , $i = 0.50 \text{ mA}$. (Vertical scale: $i_s = 2.0 \text{ mA}$ at the top.)

Resistance (before the field, only the battery drives the current):

$$R = \frac{\varepsilon}{i} = \frac{6.00 \times 10^{-6}}{1.5 \times 10^{-3}} = 4.0 \times 10^{-3} \Omega$$

During the interval, the induced emf opposes the battery (current decreases):

$$i' = \frac{\varepsilon - \varepsilon_{ind}}{R} \Rightarrow \varepsilon_{ind} = \varepsilon - i'R = 6.00 \times 10^{-6} - (0.50 \times 10^{-3})(4.0 \times 10^{-3}) = 4.0 \times 10^{-6} \text{ V}$$

From Faraday's law, with $B = at$ so $dB/dt = a$:

$$\varepsilon_{ind} = A \frac{dB}{dt} = Aa \Rightarrow a = \frac{\varepsilon_{ind}}{A} = \frac{4.0 \times 10^{-6}}{5.0 \times 10^{-4}} = \underline{\mathbf{8.0 \times 10^{-3} \text{ T/s}}}$$